Ratio's & Proportions [Vector India Pvt. Ltd]

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↳ What is Ratio?

Ratio is the representation of no of parts of two (or) more individual quantities in a mixture of same kind.

↳ Purpose of ratio?

Ratio is used to compare 2 or more individual quantities of a mixture.

Note:- Ratio is different from fraction. Both these are representations only but representing 2 (or) more individual quantities is ratio & representing one individual quantity is fraction.

Difference B/w Ratio & Fraction

Ratio	Fractions
Ex:- 2:3	Ex:- $\frac{2}{3}$
Total Quantities:- 2 individual Quantities	→ only one individual Quantity
Total No of parts:- (2+3) = 5 parts	→ 2 parts out of 3 parts means 3 parts
Individual Quantities: No of parts $Q_1 = 2$ parts, $Q_2 = 3$ parts	→ $Q_1 = 2$ parts
Individual Quantities: Values:- $Q_1 = \frac{2 \text{ parts}}{5 \text{ parts}} \times \text{Total}$	→ $Q_{value} = \frac{2}{3} \times \text{total}$
$Q_2 = \frac{3 \text{ parts}}{5 \text{ parts}} \times \text{Total}$	

Proportion:- When 2 (or) more ratio's are equal then they are said to be in proportion.

Ex:- $a : b = c : d$

When 2 ratio's are in proportion then product of extremes = product of means.

$a : b = c : d$ [or $\frac{a}{b} :: \frac{c}{d}$]

4th proportion:- When 2 ratios are in proportion such a way that $a:b = c:d$ then 'd' is called 4th propn.

$a:b = c:d \Rightarrow ad = bc$

$\Rightarrow$ $\boxed{d = 4^{th} \text{ proportion} = \frac{bc}{a}}$

Ex:- 4th proportion to 2, 3 & 4 ? [a=2, b=3, c=4.]

$\therefore d = \frac{bc}{a} = \frac{3 \times 4}{2} = 6.$

3rd & Mean (2nd) Proportions:- When 2 ratios are in proportions such a way that $a:b = b:c$ then 'c' is called 3rd proportion & 'b' is called mean proportion (or) 2nd proportion.

$a:b = b:c \Rightarrow b^2 = ac.$

$\boxed{c = \frac{b^2}{a} = 3^{rd} \text{ proportion}}$ $\boxed{b = \sqrt{ac} = \text{Mean (or) } 2^{nd} \text{ proportion}}$

↳ Duplicate ratio of $a:b$ is $a^2:b^2$

↳ Sub-Duplicate ratio of $a:b$ is $\sqrt{a}:\sqrt{b}$

↳ Triplicate ratio of $a:b$ is $a^3:b^3$

↳ Sub-triplicate ratio of $a:b$ is $\sqrt[3]{a}:\sqrt[3]{b}$

Ex:1:- $a:b = 1:2$, $b:c = 3:4$, $c:d = 5:6$ then $a:b:c:d = ?$

↳ To find compound ratio [a:b:c··· when a:b, b:c.... are given] we have to equalise the no. of parts of common terms in the ratios.

$a:b = 1:2$, $b:c = 3:4$

Equalise the no. of parts of b (common term) in both the ratio's by multiplying 1st ratio with 3 & 2nd ratio with '2'.

$a:b = [1:2]3$, $b:c = [3:4]2$

$= 3:6$ $= 6:8$

$a:b:c = 3:6:8$

(or)

a:b → 1 : 2 (×3)

b:c → 3 : 4 (×2)

a:b:c = 1×3 : 3×2 : 2×4

= 3:6:8

(or)

$\frac{a}{b} = \frac{1}{2}\frac{(n_1)}{(d_1)}$, $\frac{b}{c} = \frac{3}{4}\frac{(n_2)}{(d_2)}$

$a:b:c = 1\times3 : 3\times2 : 2\times4 = n_1n_2 : d_1n_2 : d_1d_2$

★↳ a:b:c = ratio of product of numerator to product of denominators in such a way that one by one numerator is replaced with denominator and applicable for 2 (or) more ratios are given.

Ex:2:- $a:b = \overset{n_1}{2}:\overset{d_1}{3}$, $b:c = \overset{n_2}{4}:\overset{d_2}{5}$, $c:d = \overset{n_3}{6}:\overset{d_3}{7}$, $d:e = \overset{n_4}{8}:\overset{d_4}{9}$

$a:b:c:d:e = ?$

$a:b:c:d:e = n_1n_2n_3n_4 : d_1n_2n_3n_4 : d_1d_2n_3n_4 : d_1d_2d_3n_4 : d_1d_2d_3d_4$